```c
#include <stdio.h>
#include <string.h>

int main() {
    char data[100], gen[20], code[120], temp[120];
    int n, m, i, j, error = 0;

    printf("Enter data bits: ");
    scanf("%s", data);

    printf("Enter generator bits: ");
    scanf("%s", gen);

    n = strlen(data);
    m = strlen(gen);

    strcpy(code, data);

    for (i = 0; i < m - 1; i++)
        code[n + i] = '0';
    code[n + m - 1] = '\0';

    strcpy(temp, code);

    for (i = 0; i < n; i++) {
        if (temp[i] == '1') {
            for (j = 0; j < m; j++)
                temp[i + j] =
                    (temp[i + j] == gen[j]) ? '0' : '1';
        }
    }

    for (i = 0; i < m - 1; i++)
        code[n + i] = temp[n + i];

    code[n + m - 1] = '\0';

    printf("CRC remainder: ");
    for (i = n; i < n + m - 1; i++)
        printf("%c", code[i]);

    printf("\nTransmitted code: %s\n", code);

    printf("Enter received code for verification: ");
    scanf("%s", temp);
```

```
    n = strlen(temp);

    for (i = 0; i <= n - m; i++) {
        if (temp[i] == '1') {
            for (j = 0; j < m; j++)
                temp[i + j] =
                    (temp[i + j] == gen[j]) ? '0' : '1';
        }
    }

    for (i = 0; i < n; i++) {
        if (temp[i] == '1') {
            error = 1;
            break;
        }
    }

    if (error)
        printf("Error detected!\n");
    else
        printf("No error detected.\n");

    return 0;
}
```

**Note: Enter binary strings only, and use a valid generator polynomial whose first and last bits are 1.**